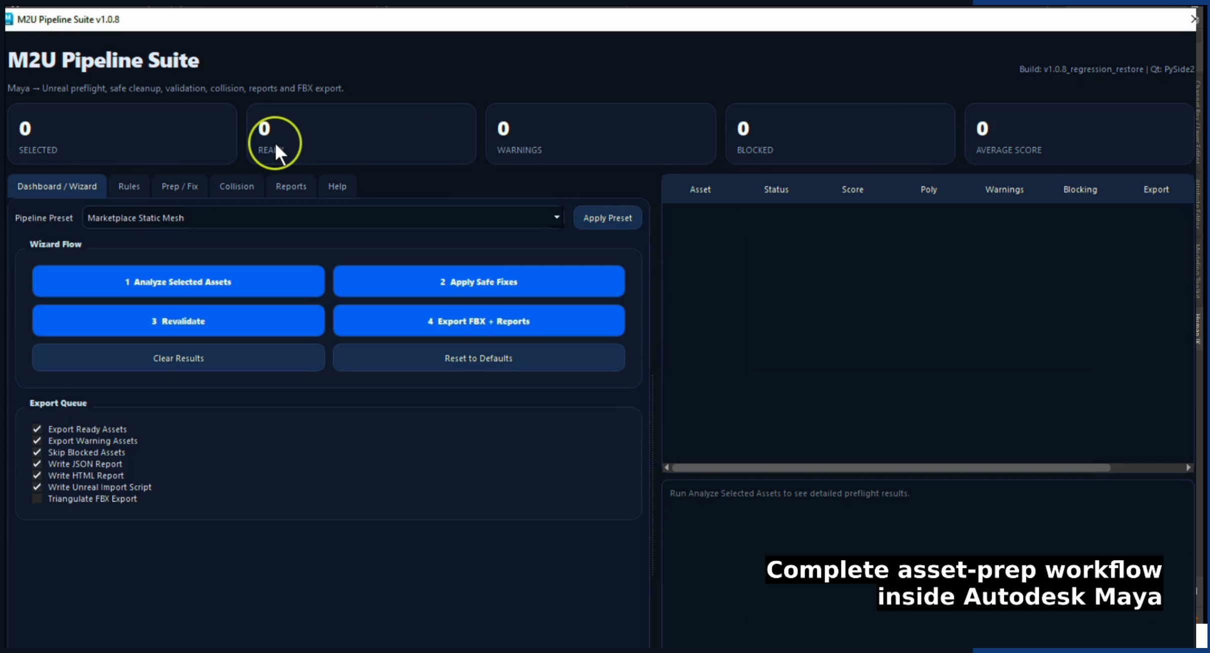
M2U Pipeline Suite v1.0.8 Professional Documentation

Maya-to-Unreal Static Mesh Preflight, Safe Cleanup, Validation, Collision, Reporting and FBX Export Toolkit



Prepared for artists, technical artists, asset creators and pipeline users working between Autodesk Maya and Unreal Engine.

Developer: Sonat Birdane | Public documentation package | Generated May 2026

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1. Executive Overview

M2U Pipeline Suite is a production-oriented Autodesk Maya toolset for preparing static mesh assets before they are imported into Unreal Engine. It combines preflight validation, artist-controlled cleanup, UCX collision matching, FBX export, report generation and Unreal import script output in one workflow.

The tool is built around a four-step workflow: analyze the current Maya selection, apply safe fixes where appropriate, revalidate the processed assets, then export FBX files and quality assurance reports. This structure keeps artists in control while removing repetitive, error-prone cleanup work from the export process.

> Core value
> M2U Pipeline Suite is not just an FBX exporter. It is a Maya-to-Unreal asset-preparation system designed to detect avoidable import problems before assets leave Maya.

Area	Purpose
Preflight validation	Find naming, transform, construction history, pivot, grid, dimension, material, UV, LOD, socket and collision issues.
Safe cleanup	Create backups, sanitize names, add prefixes, sync UCX names, freeze transforms, delete history, move pivots and optionally snap to grid.
Collision workflow	Detect Unreal-style UCX collision meshes, support required/optional/forbidden modes and include matched UCX meshes in FBX export.
Reporting	Write JSON data for pipeline use and HTML reports for artist-friendly QA review.
Unreal import support	Generate a Python import script intended to run inside Unreal Editor.

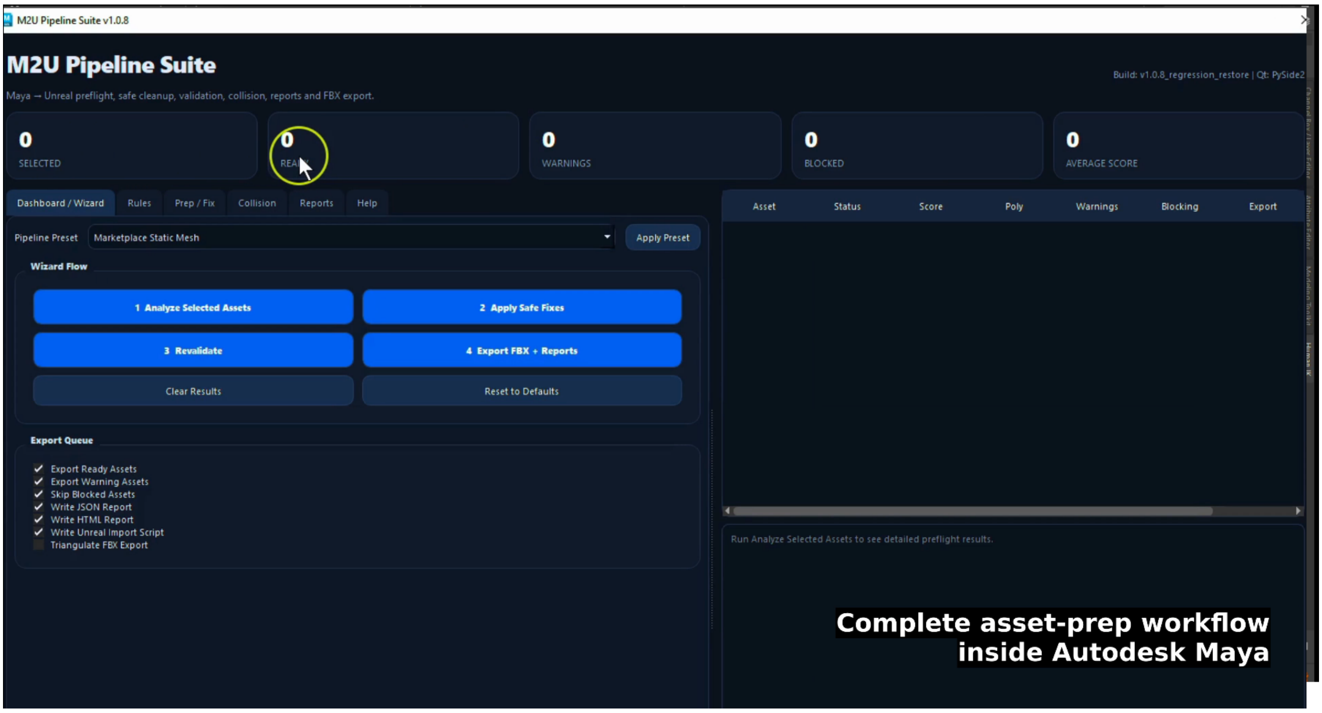


Figure 1 - Main dashboard with the complete asset-preparation workflow inside Autodesk Maya.

2. Audience and Production Scope

This documentation is intended for individual artists, technical artists, pipeline TDs, Unreal marketplace creators and small teams that need consistent Maya-to-Unreal static mesh export behavior. The terminology assumes familiarity with Maya transforms, static mesh assets, FBX export, Unreal collision naming and basic game-art production practices.

The tool is intentionally conservative. It automates safe, repeatable cleanup tasks and provides detailed reports, but it does not replace manual art direction, geometry review, UV authoring, complex collision authoring or Unreal-side final validation.

Designed for	Not designed for
Static mesh preflight and export	Skeletal mesh rigging or animation export
Marketplace asset packs, props, furniture and modular kits	Automatic mesh remodeling or topology repair
UCX collision matching and export inclusion	Advanced convex decomposition for complex collision
QA reports and pipeline transparency	Full Unreal project configuration or material creation

3. Supported Environment

The public release is targeted at Autodesk Maya 2024 and newer. The interface includes compatibility handling for PySide2 and PySide6, which helps support modern Maya versions. FBX export requires the Maya FBX plug-in.

Component	Requirement or recommendation
Autodesk Maya	Maya 2024 or newer.
Python	Maya Python 3 environment.
UI framework	PySide2 in Maya 2024, with fallback handling for PySide6 in newer versions.
FBX	Maya FBX plug-in available and loadable.
Unreal Engine	Recommended for Unreal Engine 5.x when using exported static meshes or the generated import script.
Unreal Python	Python Editor Script Plugin enabled if running the generated import script inside Unreal Editor.

4. Package Contents

A typical public release package contains the main Maya tool, a launcher, an Easy Installer, a shelf-button reinstall helper, the icon file and the documentation files.

File	Role
m2u_pipeline_suite_v1_0.py	Main Maya tool.
M2U Pipeline Suite.py	Simple launcher script for Maya.
M2U_Pipeline_Suite_v1_0_8_Easy_Installer.py	Easy Installer that copies files and creates or updates the Maya shelf button.

File	Role
reinstall_m2u_pipeline_suite_shelf_button.py	Recreates the Maya shelf button without reinstalling the whole tool.
m2u_pipeline_suite_icon.png	Custom shelf icon.
M2U_Pipeline_Suite_README.txt	General user documentation.
M2U_Pipeline_Suite_FUNCTIONS.txt	Detailed function and option reference.

5. Installation and Launch

Recommended installation

- Extract the package to a temporary location.
- Open Autodesk Maya.
- Open Windows > General Editors > Script Editor.
- Switch to the Python tab.
- Open or drag the Easy Installer script into the Python tab.
- Run M2U_Pipeline_Suite_v1_0_8_Easy_Installer.py.
- Choose the folder where the tool should be installed.
- The installer copies the required files and creates or updates the M2U_Tools shelf.
- Launch the tool from Maya Shelf > M2U_Tools > M2U Pipeline Suite.

> After installation
> Users do not need to run the installer again. The tool can be launched directly from the M2U_Tools shelf. If the shelf button is removed or Maya preferences are reset, run the shelf-button reinstall script.

Launching the tool

After installation, use the shelf button named M2U Pipeline Suite under the M2U_Tools shelf. If the icon is missing, confirm that the icon file is installed next to the tool files and rerun the shelf-button reinstall helper.

6. Quick Start Workflow

The tool is optimized around a simple loop that artists can repeat for individual assets or batches of selected assets.

Step	Action	Result
1	Select one or more base static mesh transforms in Maya.	The selected transforms become the input scope.
2	Open M2U Pipeline Suite and choose a preset.	The UI is configured for a common production workflow.

Step	Action	Result
3	Click Analyze Selected Assets.	The dashboard, asset table and detail panel show readiness results.
4	Review warnings, blocking issues and the fix plan.	The artist decides whether the proposed cleanup is appropriate.
5	Click Apply Safe Fixes when acceptable.	Enabled cleanup operations run, with backups if configured.
6	Click Revalidate.	The tool confirms the current scene state after cleanup.
7	Click Export FBX + Reports.	Eligible assets are exported and reports/import scripts are written.
8	Open the HTML report.	The final export status can be reviewed or shared.

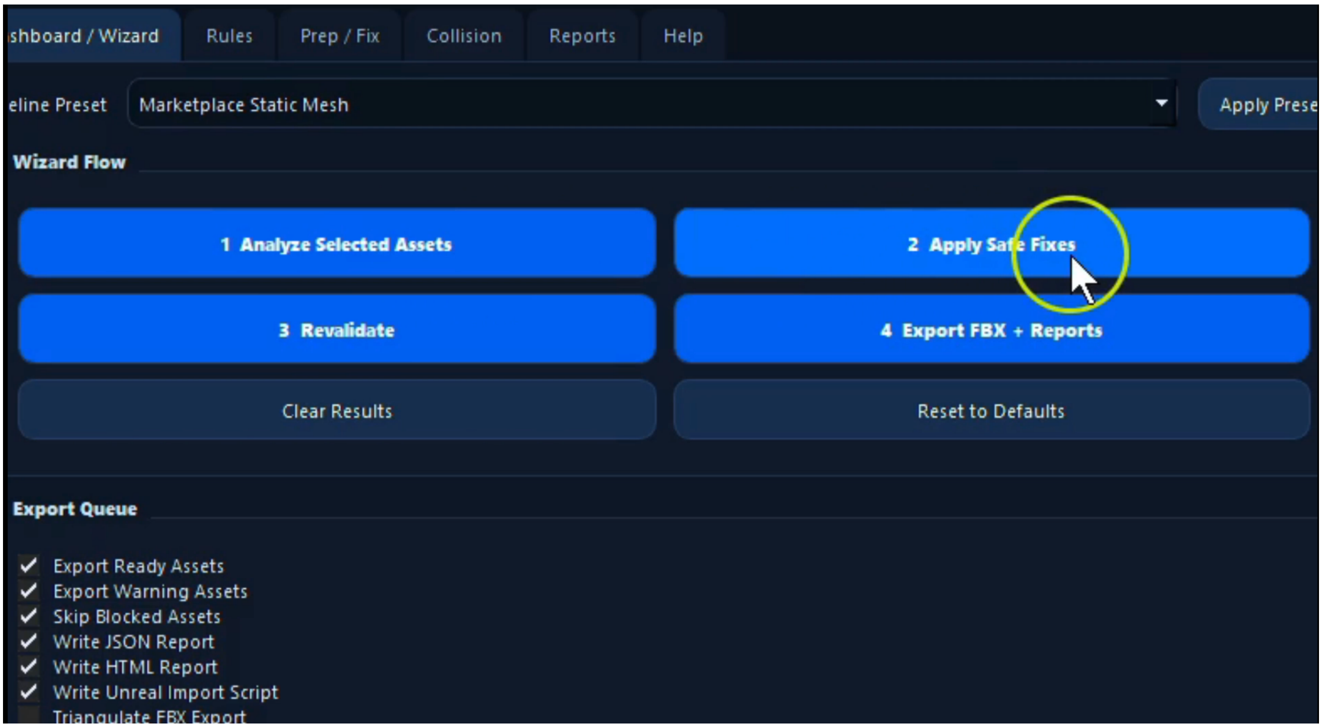


Figure 2 - Wizard Flow buttons: Analyze, Apply Safe Fixes, Revalidate and Export FBX + Reports.

7. Interface Overview

Dashboard / Wizard

The Dashboard / Wizard tab is the primary working area. It contains the pipeline preset selector, the four workflow buttons, export queue settings, dashboard summary cards, the asset result table and the detail panel.

Rules

The Rules tab defines what the tool should check: naming, geometry, dimensions, pivot targets, grid readiness and advanced Unreal-readiness categories. Rule severities can be configured as off, warning or blocking depending on the desired strictness.

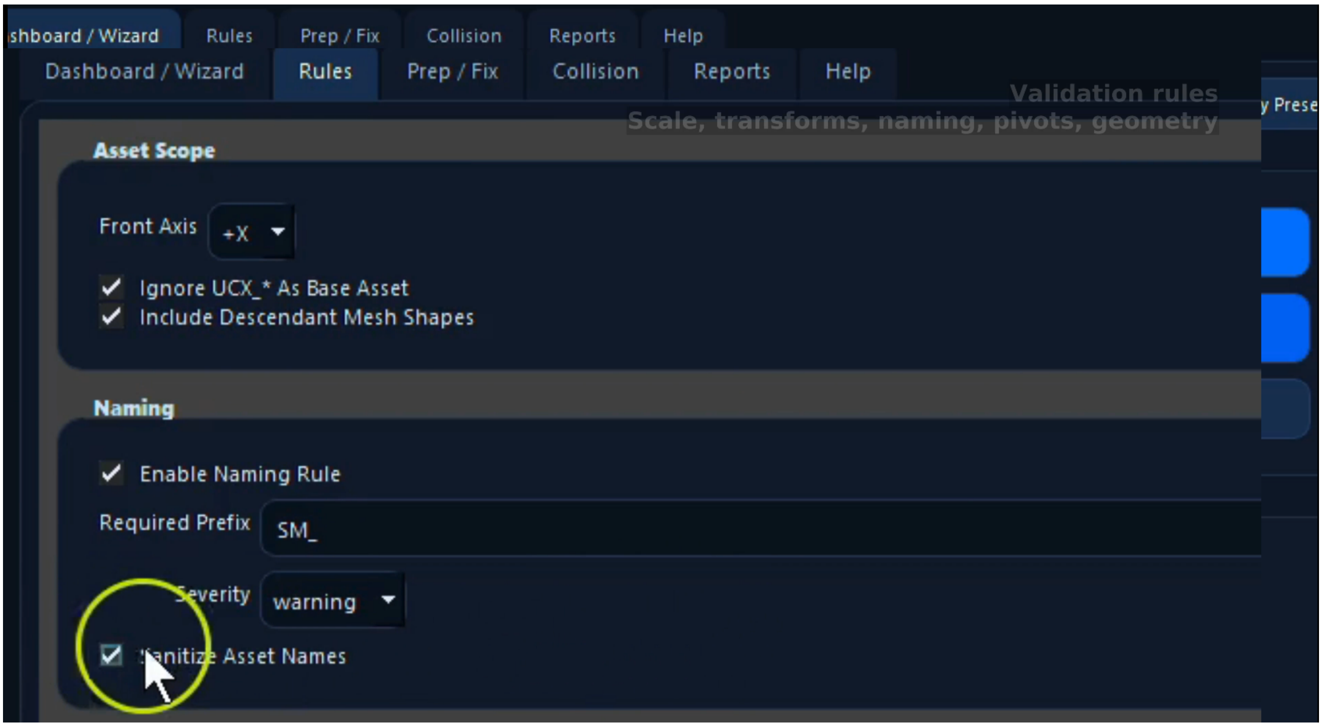


Figure 3 - Rules tab showing asset scope, naming validation and sanitization settings.

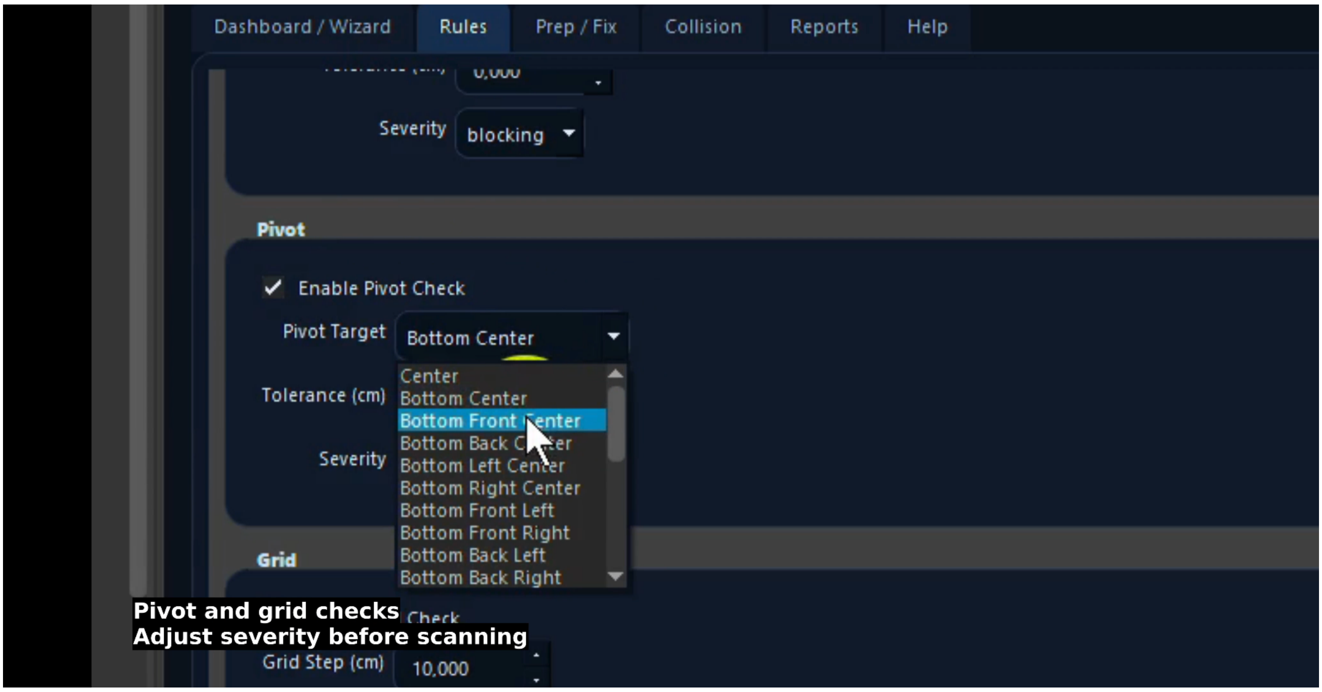


Figure 4 - Pivot target configuration with front/back/left/right targets calculated from the selected Front Axis.

Prep / Fix

The Prep / Fix tab contains the safe-fix actions that can modify scene objects. It also contains artist-controlled preview helpers such as pivot preview locators and simple box UCX creation.

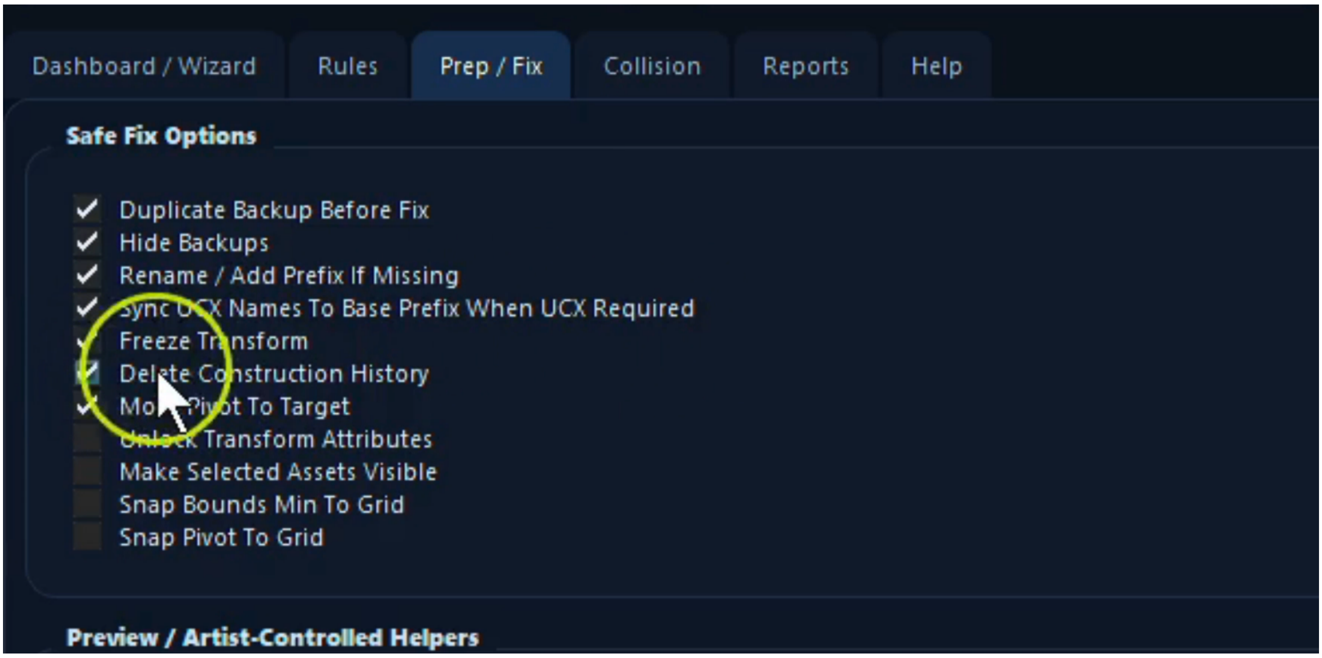


Figure 5 - Safe Fix options for backups, rename, UCX sync, transform cleanup and pivot/grid operations.

Collision

The Collision tab configures UCX requirements, match mode, multiple-part acceptance, matched UCX validation and collision visualizer helpers.

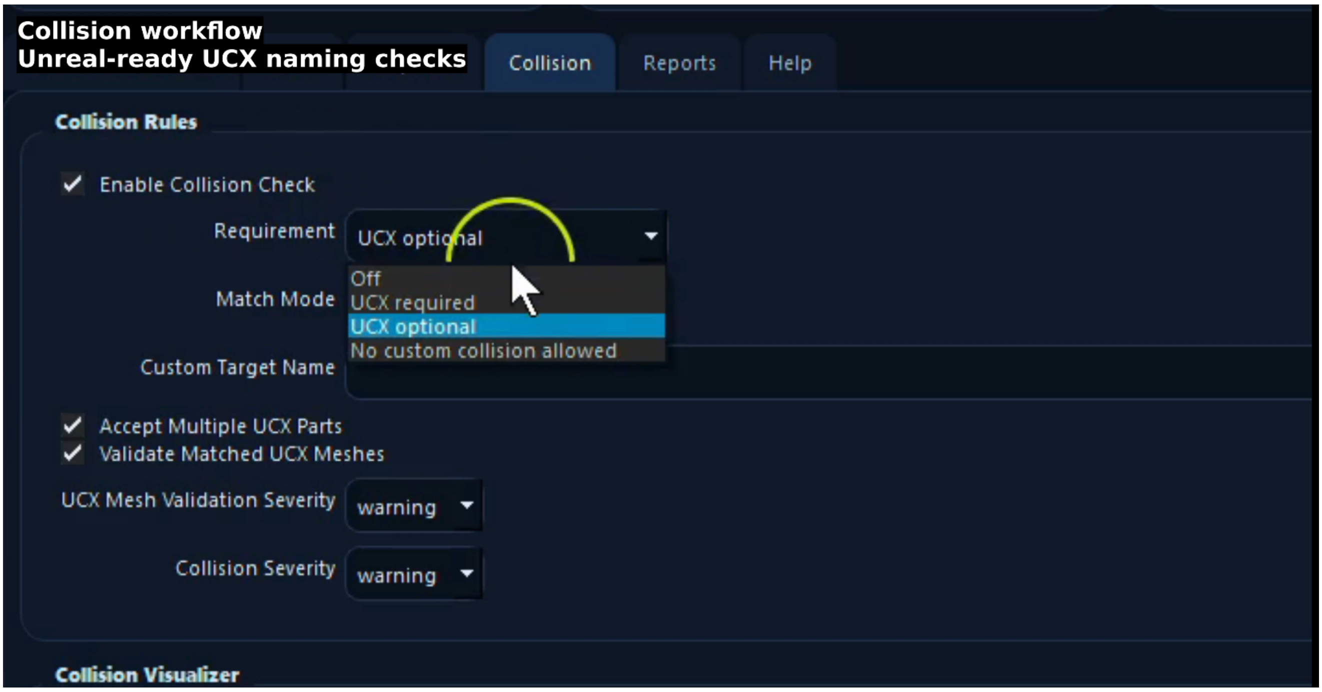


Figure 6 - Collision workflow with Unreal-ready UCX naming checks and requirement modes.

Reports

The Reports output includes a structured JSON report, an HTML QA report and an optional Unreal Python import script. The HTML report is intended for human review and delivery documentation.

8. Presets

Presets provide a fast starting point. Selecting a preset from the dropdown does not change settings until Apply Preset is clicked. After applying a preset, users can still adjust individual rules manually.

Preset	Best for	Typical behavior
Marketplace Static Mesh	General props, furniture, environment pieces and marketplace assets.	SM_ prefix, UCX optional, freeze required, clean history warning, bottom-center pivot, grid warning and reports enabled.
Modular Wall Kit	Grid-based modular walls, floors, trims, doors, windows and kit pieces.	UCX required, strict grid readiness, dimension checks enabled and placement-focused pivot targets.
Furniture / Prop	Standalone furniture and decoration props.	Flexible polycount, UCX optional, bottom-center pivot and material/UV readiness checks.
Collision Strict	Assets that must ship with custom collision.	UCX required, collision severity blocking and stricter matched UCX validation.
Nanite Candidate Check	High-detail static mesh candidates.	Higher polycount threshold, Nanite-oriented review and reduced emphasis on classic low-poly constraints.

9. Status, Severity and M2U Score

Asset statuses

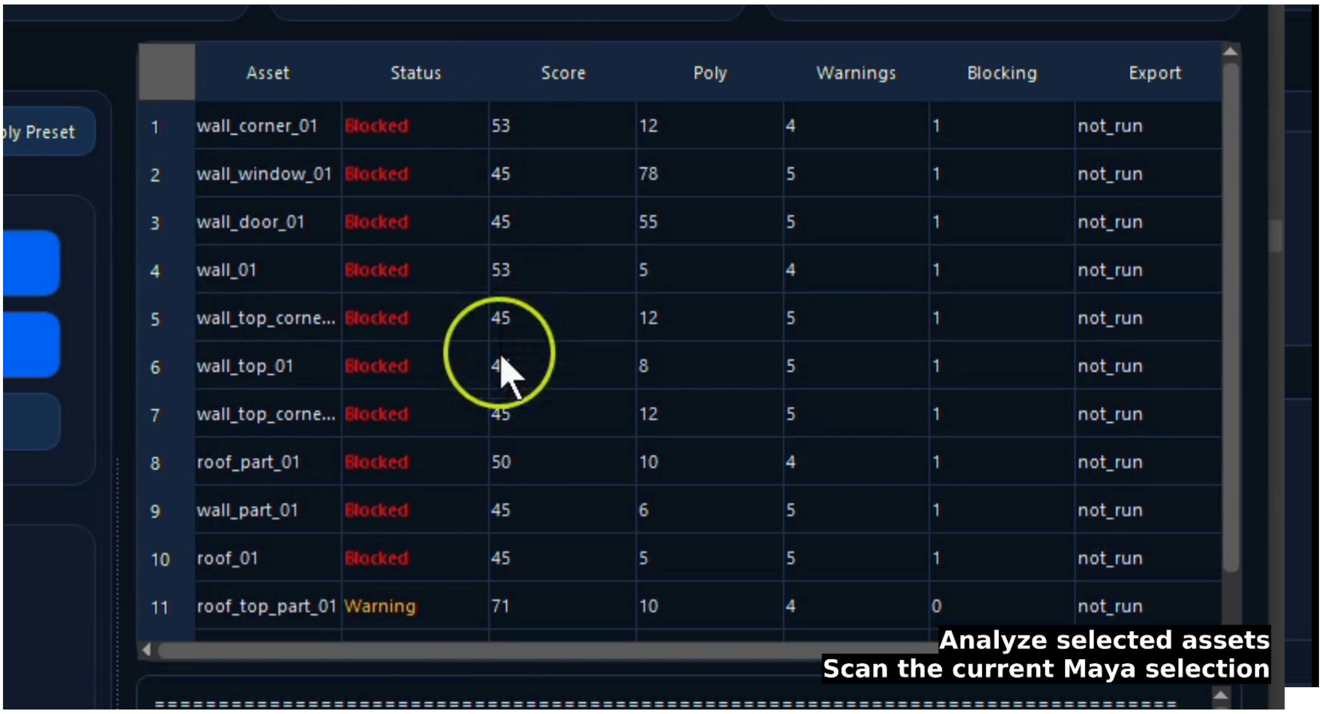
Status	Meaning	Export behavior
Ready	No blocking issues were found.	Normally exported when Export Ready Assets is enabled.
Warning	Warnings exist, but no blocking issues were found.	Exported only when Export Warning Assets is enabled.
Blocked	One or more blocking issues were found.	Skipped by default when Skip Blocked Assets is enabled.

Severity levels

Severity	Behavior
Off	The rule does not create warnings or blocking issues. Use this when a rule is not relevant to the current workflow.
Warning	The rule can fail but the asset can still be exported if warning exports are enabled.
Blocking	The rule can fail and block export. Use this for requirements that must be fixed before delivery.

M2U Score

Each asset receives an M2U Score from 0 to 100. The score starts at 100 and is reduced by warnings, blocking issues and failed checks. It is a quick quality indicator, not a replacement for reading the detailed results.



	Asset	Status	Score	Poly	Warnings	Blocking	Export
1	wall_corner_01	Blocked	53	12	4	1	not_run
2	wall_window_01	Blocked	45	78	5	1	not_run
3	wall_door_01	Blocked	45	55	5	1	not_run
4	wall_01	Blocked	53	5	4	1	not_run
5	wall_top_corne...	Blocked	45	12	5	1	not_run
6	wall_top_01	Blocked	4	8	5	1	not_run
7	wall_top_corne...	Blocked	45	12	5	1	not_run
8	roof_part_01	Blocked	50	10	4	1	not_run
9	wall_part_01	Blocked	45	6	5	1	not_run
10	roof_01	Blocked	45	5	5	1	not_run
11	roof_top_part_01	Warning	71	10	4	0	not_run

Figure 7 - Analysis results table showing per-asset status, score, polycount, warnings, blocking issues and export state.

10. Analyze, Fix, Revalidate and Export

Analyze Selected Assets

Analyze Selected Assets inspects selected Maya base mesh transforms and creates a validation result for each asset. It does not modify the Maya scene. The result includes asset status, M2U Score, checks, warnings, blocking issues, fix plan, collision matches and advanced readiness results.

Apply Safe Fixes

Apply Safe Fixes runs only the enabled cleanup operations. Depending on the options, it can create hidden backup duplicates, rename/sanitize base mesh names, add missing prefixes, sync matching UCX names, delete construction history, freeze transforms, move pivots, fix visibility, unlock transform attributes and optionally snap to grid.

After Safe Fixes, the tool restores Maya selection to the processed real base assets rather than leaving backup objects selected. This allows the user to continue directly to Revalidate or Export without manually reselecting assets.

Revalidate

Revalidate runs the analysis again on the current scene state. It is recommended after any cleanup operation because it confirms what actually changed and whether warnings or blocking issues remain.

Figure 8 - Revalidation after cleanup: assets are no longer blocked, but warnings remain for issues that require review.

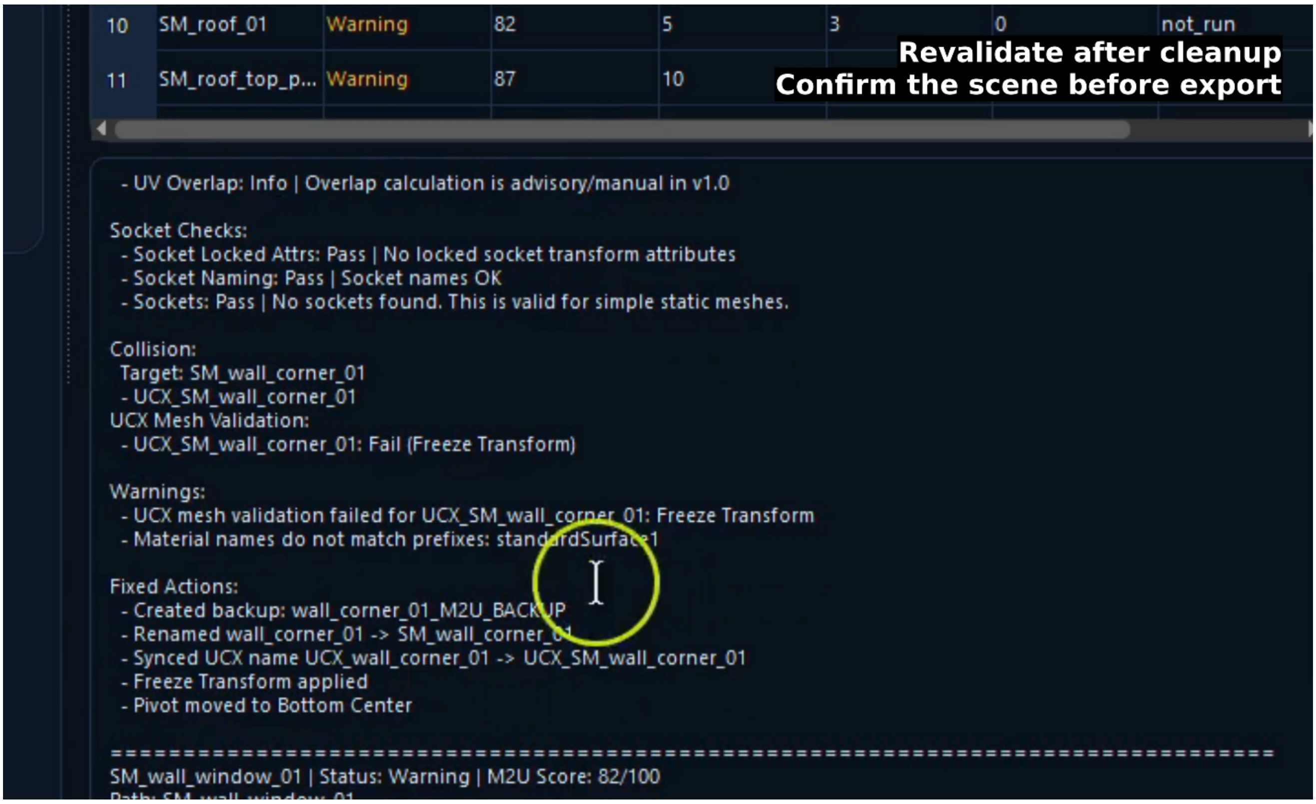


Figure 9 - Detail panel listing warnings, collision matches and fixed actions after cleanup.

Export FBX + Reports

Export FBX + Reports exports eligible assets to FBX and writes selected reports. Before exporting each asset, the tool performs a fresh UCX search so it does not rely on stale analysis results. This matters when UCX meshes were created, renamed or synchronized after the first analysis.

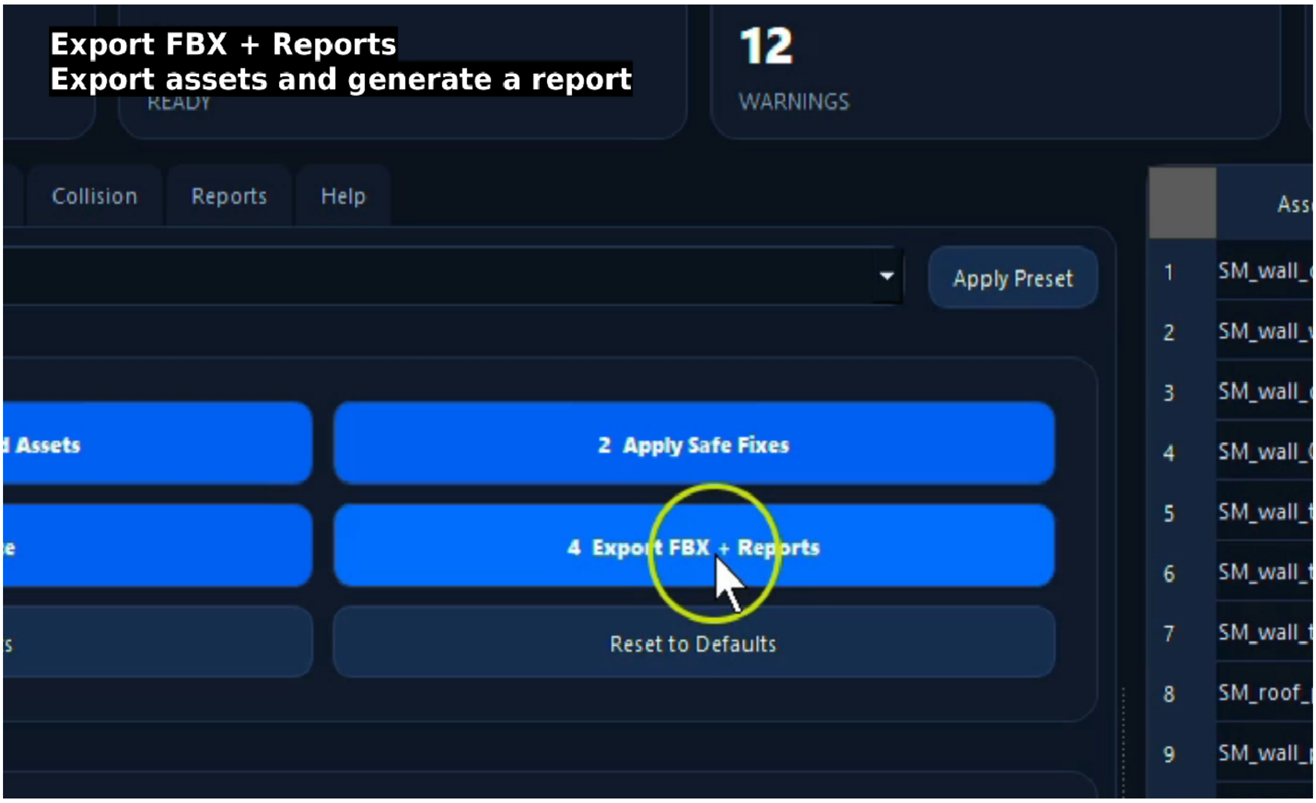


Figure 10 - Export FBX + Reports generates asset exports and documentation from the Wizard workflow.

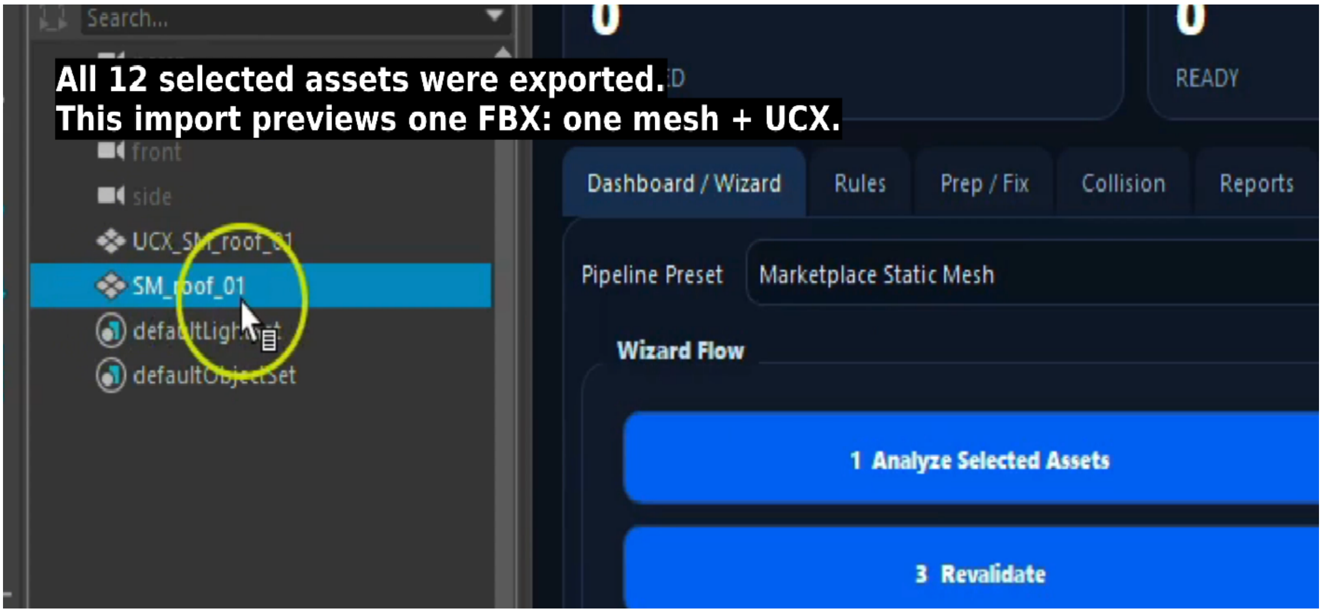


Figure 11 - Export result preview showing one FBX containing one base mesh and its matching UCX collision.

11. UCX Collision Workflow

The UCX collision system follows Unreal-style naming. A base mesh named SM_Table_01 should normally be matched by UCX_SM_Table_01, UCX_SM_Table_01_01 or similar numbered multipart collision meshes.

Collision requirement	Description
Off	Collision checks are ignored.

Collision requirement	Description
UCX required	A matching UCX mesh must exist. Missing UCX can warn or block, and export-time missing UCX fails that asset.
UCX optional	Missing UCX is acceptable, but matching UCX is included and can still be validated.
No custom collision allowed	Matching UCX is considered a failure. Use this when custom collision should not be exported.

UCX prefix sync

When Safe Fix adds a required base mesh prefix, the tool can also rename matching UCX collision meshes so the collision target remains valid. Multipart UCX suffixes are preserved.

```
Before Safe Fix:
Chair_01
UCX_Chair_01
UCX_Chair_01_02

Required Prefix:
SM_

After Safe Fix:
SM_Chair_01
UCX_SM_Chair_01
UCX_SM_Chair_01_02
```

Export-time UCX recheck

Before exporting an asset, the tool searches for UCX matches again. If UCX Required is active and no matching UCX is found at export time, export fails for that asset instead of exporting the base mesh alone.

12. Reporting System

M2U Pipeline Suite can generate both technical and artist-friendly reports. JSON is useful for pipeline integration, debugging and structured data review. HTML is designed for visual QA, support, delivery documentation and team review.

Output	Purpose	Typical contents
M2U_Pipeline_Report.json	Machine-readable report.	Settings, export summary, per-asset results, warnings, blocking issues, fix actions, collision matches and validation data.
M2U_Pipeline_Report.html	Human-readable QA report.	Summary cards, per-asset status, score, issues, fix plan, fixed actions, core checks and readiness checks.
M2U_Unreal_Import_Assets.py	Unreal import helper.	Automated static mesh import tasks for exported FBX files.

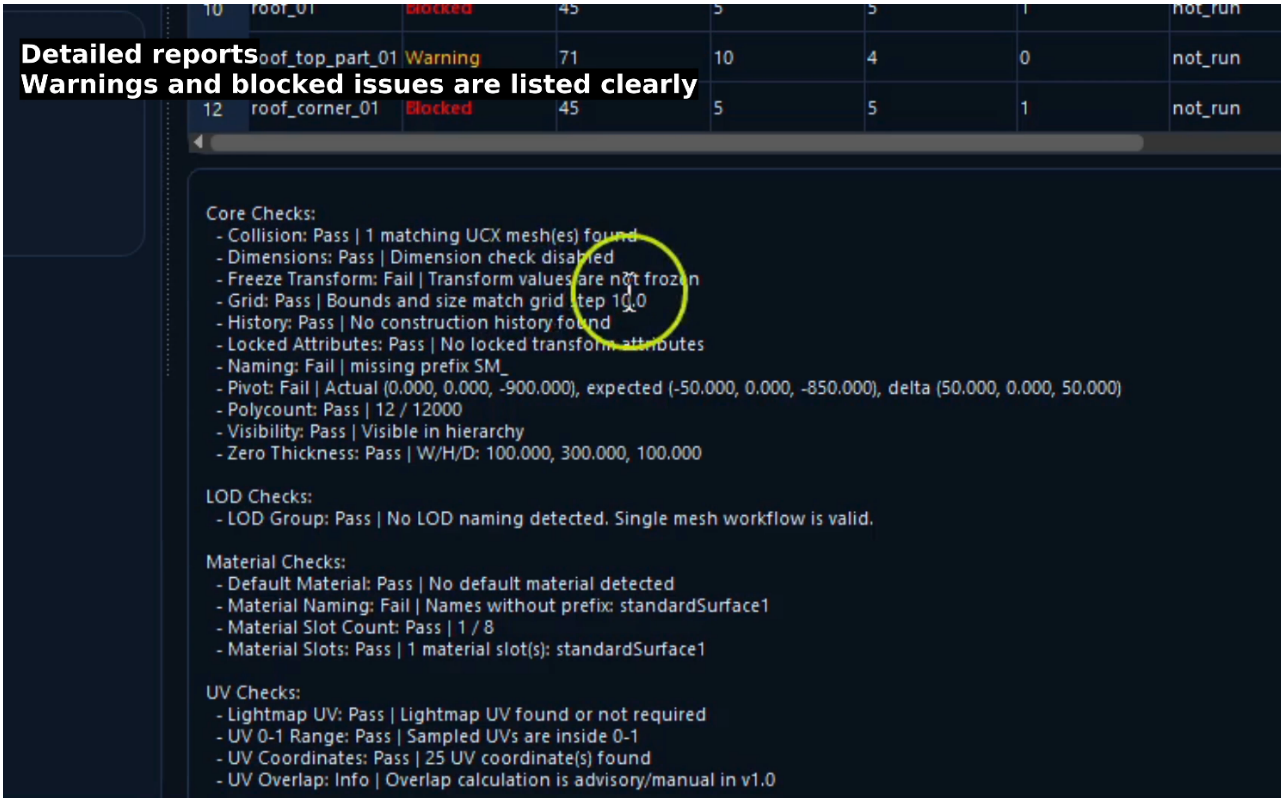


Figure 12 - Detail view listing core checks, warnings and blocked issues clearly.

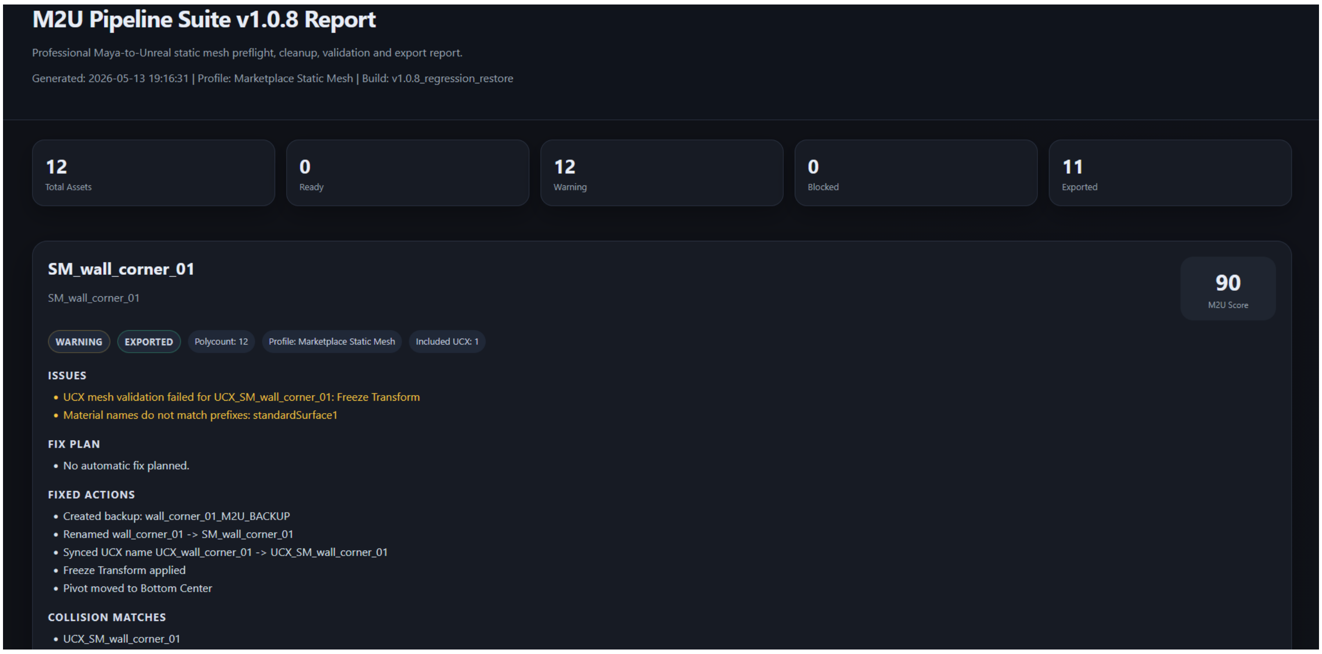
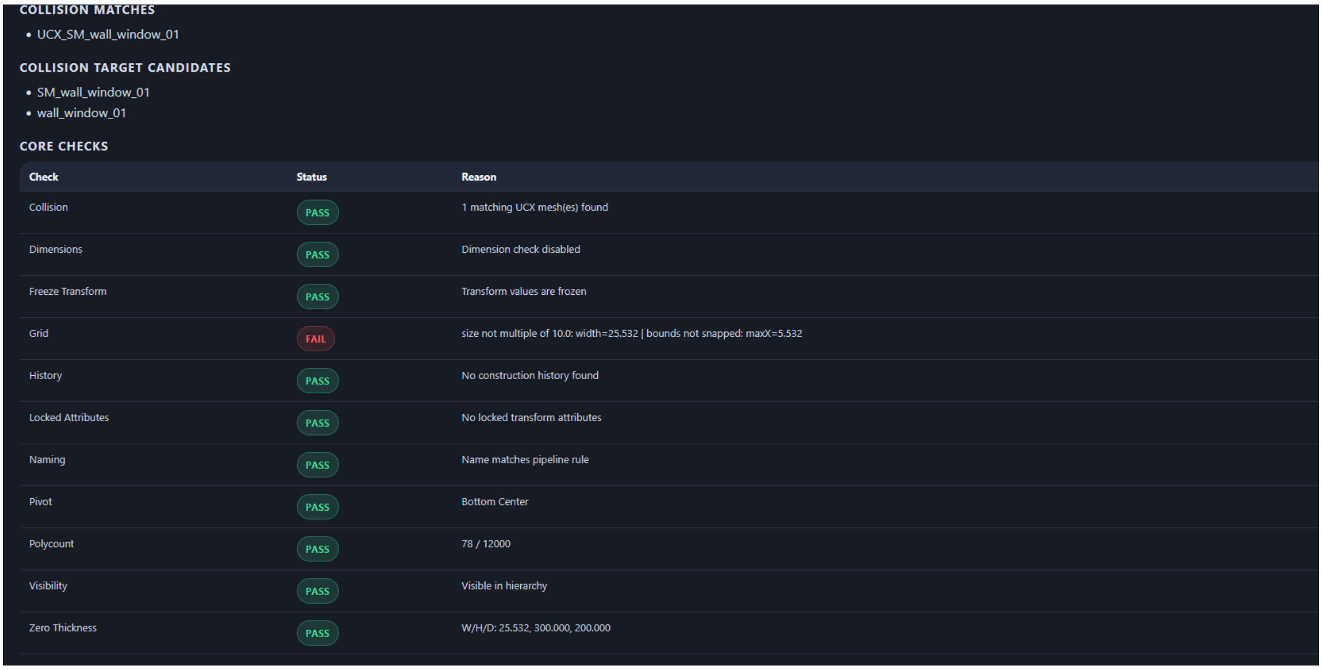


Figure 13 - HTML report overview with summary cards, asset status, score, issues, fixed actions and collision matches.



COLLISION MATCHES

- UCX_SM_wall_window_01

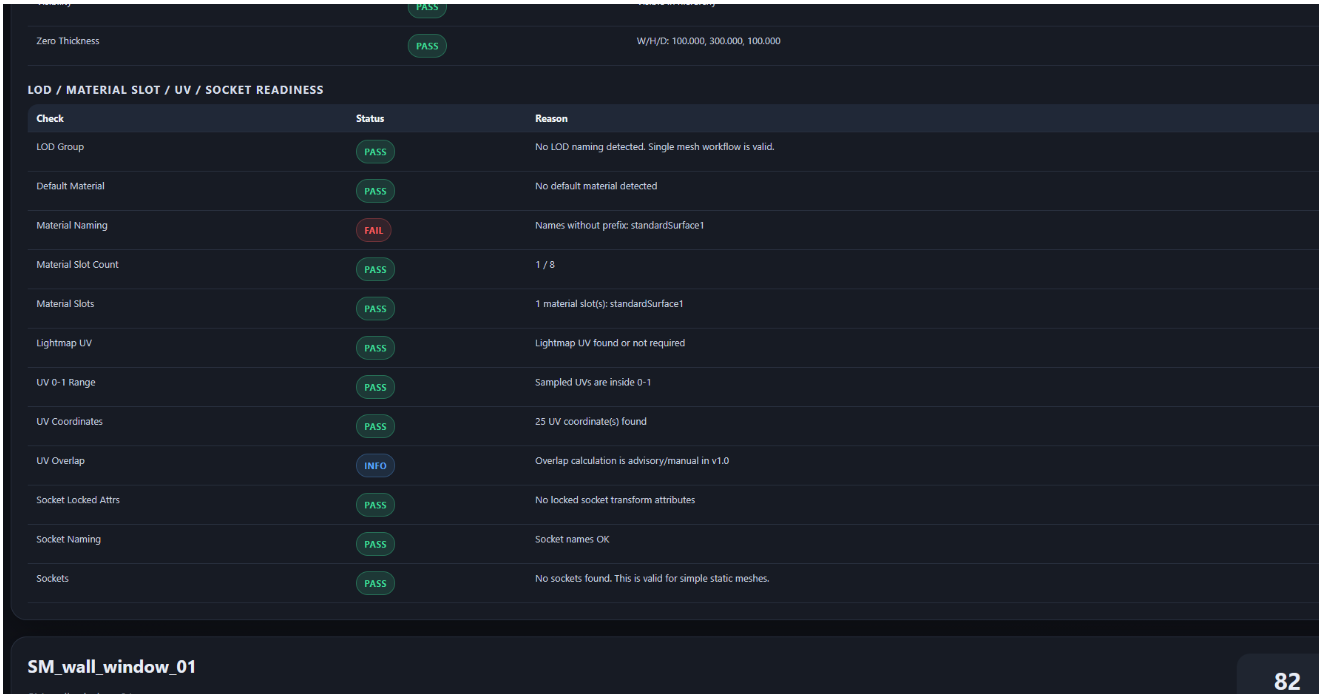
COLLISION TARGET CANDIDATES

- SM_wall_window_01
- wall_window_01

CORE CHECKS

Check	Status	Reason
Collision	PASS	1 matching UCX mesh(es) found
Dimensions	PASS	Dimension check disabled
Freeze Transform	PASS	Transform values are frozen
Grid	FAIL	size not multiple of 10.0: width=25.532 | bounds not snapped: maxX=5.532
History	PASS	No construction history found
Locked Attributes	PASS	No locked transform attributes
Naming	PASS	Name matches pipeline rule
Pivot	PASS	Bottom Center
Polycount	PASS	78 / 12000
Visibility	PASS	Visible in hierarchy
Zero Thickness	PASS	W/H/D: 25.532, 300.000, 200.000

Figure 14 - HTML report core checks table with pass/fail status and reason columns.



Zero Thickness	PASS	W/H/D: 100.000, 300.000, 100.000

LOD / MATERIAL SLOT / UV / SOCKET READINESS

Check	Status	Reason
LOD Group	PASS	No LOD naming detected. Single mesh workflow is valid.
Default Material	PASS	No default material detected
Material Naming	FAIL	Names without prefix: standardSurface1
Material Slot Count	PASS	1 / 8
Material Slots	PASS	1 material slot(s): standardSurface1
Lightmap UV	PASS	Lightmap UV found or not required
UV 0-1 Range	PASS	Sampled UVs are inside 0-1
UV Coordinates	PASS	25 UV coordinate(s) found
UV Overlap	INFO	Overlap calculation is advisory/manual in v1.0
Socket Locked Attrs	PASS	No locked socket transform attributes
Socket Naming	PASS	Socket names OK
Sockets	PASS	No sockets found. This is valid for simple static meshes.

SM_wall_window_01

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Figure 15 - HTML report readiness checks for LOD, material slots, UVs and sockets.

13. Unreal Import Script

The generated Unreal import script is intended to run inside Unreal Editor, not inside Maya. It imports the exported FBX files into a default Unreal content destination such as /Game/M2U_Imported.

The generated script configures automated static mesh import tasks with mesh import enabled, skeletal import disabled, mesh combining disabled, lightmap UV generation enabled, replace-existing enabled and save enabled. Users can edit the script if they need a different destination path or project-specific import behavior.

```
Unreal usage overview:
```

```
1. Open the Unreal project.
2. Enable Python Editor Script Plugin if needed.
3. Open Window > Output Log.
4. Switch the command input to Python mode.
5. Run:
   exec(open("PATH_TO_SCRIPT").read())
```

14. Recommended Naming Conventions

Asset type	Recommended pattern	Examples
Static mesh	SM_<AssetName>	SM_Table_01, SM_Chair_01, SM_Wall_400x300
UCX collision	UCX_<BaseMeshName> or UCX_<BaseMeshName>_01	UCX_SM_Table_01, UCX_SM_Table_01_01
Materials	M_, MI_ or MAT_ prefix	M_Wood, MI_Table_Surface, MAT_Chair_Fabric
LOD meshes	<BaseName>_LOD0, <BaseName>_LOD1	SM_Table_01_LOD0, SM_Table_01_LOD1
Sockets / locators	SOCKET_ or Socket_ prefix	SOCKET_Handle, SOCKET_VFX_Smoke, Socket_DoorPivot

15. Best Practices

- Use clear, consistent names before export and keep base mesh names aligned with UCX collision names.
- Use presets as starting points, then adjust severities based on project requirements.
- Analyze before applying fixes, and review the Fix Plan before allowing scene changes.
- Keep Duplicate Backup Before Fix enabled during cleanup, especially in production scenes.
- Use UCX Required for assets that must ship with custom collision, and UCX Optional for general props.
- Use Modular Wall Kit or Collision Strict presets for grid-based modular kits and collision-critical assets.
- Use Pivot Preview before moving pivots on placement-critical assets.
- Revalidate after any fix pass and inspect the HTML report after export.
- Test the tool on duplicate scenes or non-critical assets before using it on production work.

16. Troubleshooting

Problem	Likely cause	Recommended action
The tool does not open	Incorrect install location, missing files or script run outside Maya Python.	Confirm package files, run in Maya Python and reinstall the shelf button.
FBX export fails	FBX plug-in unavailable, blocked assets, unwritable export folder or stale scene state.	Load/check the FBX plug-in, revalidate, review blocking issues and confirm folder permissions.

Problem	Likely cause	Recommended action
UCX is not found	Name mismatch, no mesh shape, wrong match mode or stale analysis.	Check UCX naming, rerun analysis and use UCX Prefix Sync after base mesh renames.
UCX does not export	Collision requirement off, no match, blocked asset or warning export disabled.	Enable the proper collision mode, revalidate and confirm export queue options.
Material naming fails	Default Maya material name still assigned.	Rename standardSurface1 or similar names to M_, MI_ or MAT_ conventions.
Pivot result seems wrong	Front Axis does not match the asset orientation.	Adjust Front Axis and use Pivot Preview to inspect the target.
Shelf icon is missing	Icon file not installed beside tool files or shelf metadata reset.	Confirm m2u_pipeline_suite_icon.png location and rerun shelf-button reinstall helper.

17. Limitations and Intentional Boundaries

M2U Pipeline Suite is designed to be predictable and safe. It validates common pipeline problems and automates conservative cleanup, but certain tasks remain manual or Unreal-side by design.

The tool does	The tool does not
Detect common pipeline problems before export.	Automatically remodel or resize geometry.
Apply safe, artist-controlled cleanup operations.	Solve full UV packing or UV overlap problems.
Export base assets with matching UCX collision.	Create advanced convex decomposition collision.
Generate JSON/HTML reports and an Unreal Python import helper.	Create final Unreal materials, material instances, Nanite settings or complete LOD setup.
Help make asset export more transparent and repeatable.	Replace manual art review or Unreal Editor validation.

18. Detailed Function Reference

The following reference consolidates the main options and behaviors available in M2U Pipeline Suite v1.0.8. It is designed for users who need to know exactly what each control does and how it affects validation, fixing or export.

Dashboard / Wizard Tab

The Dashboard / Wizard tab is the main workflow area.

Pipeline Preset

The preset dropdown selects a predefined collection of settings. The preset is not applied until the user clicks Apply Preset.

Available presets

- Marketplace Static Mesh
- Modular Wall Kit
- Furniture / Prop
- Collision Strict
- Nanite Candidate Check

Changing the preset dropdown alone does not change settings. Clicking Apply Preset writes that preset's values into the UI.

Apply Preset

Applies the currently selected preset to the UI controls.

This can change

- Naming requirements.
- Polycount limits.
- Collision requirement.
- Grid step and severity.
- Dimension rules.
- Pivot target and severity.
- Advanced readiness checks.
- Export behavior.
- Safe-fix defaults.

Apply Preset does not modify Maya scene objects.

1 Analyze Selected Assets

Inspects the selected base mesh transforms in Maya.

It produces

- Asset status.
- M2U Score.
- Validation checks.
- Warnings.
- Blocking issues.
- Fix plan.
- Manual review notes.
- Collision matches.
- Advanced readiness results.

Selection behavior

- Selected transform nodes are inspected.
- UCX_* transforms are ignored as base assets when Ignore UCX_* As Base Asset is enabled.

- Mesh shapes under selected transforms are included when Include Descendant Mesh Shapes is enabled.

Analyze does not modify the Maya scene.

2 Apply Safe Fixes

Applies enabled cleanup operations to the current analysis results.

Possible actions

- Create backup duplicate.
- Hide backup duplicate.
- Unlock transform attributes if enabled.
- Make selected assets visible if enabled.
- Rename/sanitize base mesh name.
- Add required prefix if missing.
- Sync matching UCX names to the renamed base asset when allowed.
- Delete construction history.
- Freeze transforms.
- Move pivot to target.
- Snap bounds min to grid.
- Snap pivot to grid.
- Revalidate each processed asset after fixes.

After Apply Safe Fixes, the tool restores Maya selection to the processed real base assets rather than leaving backup objects selected.

Apply Safe Fixes can modify Maya scene objects.

3 Revalidate

Runs Analyze Selected Assets again.

Use this after Safe Fix to confirm the current asset state.

Revalidate does not directly modify Maya scene objects.

4 Export FBX + Reports

Exports eligible assets to FBX and writes selected reports.

Export performs a fresh UCX search at export time. This is important because the user may have created, renamed, or synced UCX meshes after the first analysis.

Export can create

- FBX files.
- M2U_Pipeline_Report.json.
- M2U_Pipeline_Report.html.
- M2U_Unreal_Import_Assets.py.

Export modifies files on disk but does not intentionally modify mesh geometry.

Clear Results

Clears displayed tool results.

It clears

- Dashboard cards.
- Asset result table.
- Detail panel.
- Report path panel.
- Internal last analysis/export summary.

It does not

- Rename assets.
- Delete assets.
- Delete backups.
- Change selection.
- Export files.
- Modify scene objects.

Use this when the panel contains old analysis or export information and the user wants a clean view.

Reset to Defaults

Restores friendly Unreal-ready default settings and clears displayed results.

Default behavior is based on the Marketplace Static Mesh preset.

It restores settings such as

- Required Prefix: SM_
- Naming Severity: warning
- Collision Requirement: UCX optional
- Existing UCX validation enabled
- Max Polycount: 12000
- Freeze Required: enabled, blocking
- Clean History Required: enabled, warning
- Pivot Target: Bottom Center
- Grid Step: 10 cm
- Dimension Check: disabled
- LOD / socket / material / UV readiness checks: warning
- Backup Before Fix: enabled
- Rename / sanitize / freeze / history / pivot fix: enabled
- UCX name sync: enabled
- JSON report: enabled

- HTML report: enabled
- Unreal import script: enabled

It does not modify Maya scene objects.

Export Queue Options

Export Ready Assets

When enabled, assets with Ready status are exported.

When disabled, Ready assets are skipped even if they have no issues.

Export Warning Assets

When enabled, assets with Warning status are exported.

When disabled, assets with Warning status are skipped.

This is useful when a team wants a stricter export pass but does not want to mark every warning rule as blocking.

Skip Blocked Assets

When enabled, assets with Blocked status are skipped.

When disabled, blocked assets may still be exported depending on export logic and rule state. This is generally not recommended for production use.

Important exception

If UCX Required is active and no matching UCX collision is found at export time, the export is failed for that asset even if other export options are permissive. This prevents exporting a base mesh alone when required collision is missing.

Write JSON Report

Writes M2U_Pipeline_Report.json.

The JSON report includes

- Tool name and build.
- Timestamp.
- Current settings.
- Export summary.
- Full per-asset analysis data.
- Checks, warnings, blocking issues, fix actions, collision matches and export status.

Use JSON for technical debugging or pipeline integration.

Write HTML Report

Writes M2U_Pipeline_Report.html.

The HTML report includes

- Summary cards.
- Per-asset status.
- M2U Score.
- Export status.
- Polycount.
- Included UCX count.
- Issues.
- Fix plan.
- Fixed actions.
- Collision matches.
- Collision target candidates.
- Core checks.
- LOD / Material Slot / UV / Socket readiness checks.

Use HTML for artist-friendly QA review.

Write UNREAL Import Script

Writes M2U_Unreal_Import_Assets.py.

This script is run inside Unreal Editor Python, not inside Maya.

It imports exported FBX files into

/Game/M2U_Imported

The generated script sets up automated static mesh import tasks with

- import_mesh = True
- import_as_skeletal = False
- combine_meshes = False
- generate_lightmap_u_vs = True
- replace_existing = True
- save = True

Triangulate FBX Export

When enabled, the FBX exporter is asked to triangulate exported geometry.

When disabled, the tool does not request triangulation during FBX export.

Use enabled when a pipeline requires triangulated meshes before import. Use disabled when the artist wants to preserve the authored polygon structure as much as the FBX exporter allows.

Rules Tab - Asset Scope

Front Axis

Available values

- +X
- -X
- +Z
- -Z

Front Axis controls how front/back/left/right pivot targets are calculated.

For example, Bottom Front Center means

- The bottom of the bounding box on Maya Y.
- The front side determined by Front Axis.
- The center along the remaining horizontal axis.

Changing Front Axis affects pivot validation, pivot ghost preview and pivot fix target location.

Ignore Ucx_* As Base Asset

When enabled, selected transforms beginning with UCX_ are ignored as base assets during analysis.

This prevents collision meshes from being analyzed and exported as separate render assets.

When disabled, UCX_* nodes can be treated as regular selected assets if they contain mesh shapes. This is rarely recommended.

Include Descendant Mesh Shapes

When enabled, mesh shapes under selected transform hierarchies are included.

Use enabled when a selected group or root transform represents one asset with child mesh parts.

When disabled, only direct mesh shapes under the selected transform are considered.

This affects polycount, material collection, UV checks, mesh validity and export preparation.

Rules Tab - Naming

Enable Naming Rule

When enabled, asset names are checked against the required prefix and sanitization rules.

When disabled, naming validation is skipped.

Required Prefix

Default

```
SM_
```

The tool checks whether each base asset name starts with this prefix.

Examples

SM_Table_01 passes when required prefix is SM_. Table_01 fails when required prefix is SM_. ZM_Table_01 passes when required prefix is ZM_.

This prefix is also used by Safe Fix when Rename / Add Prefix If Missing is enabled.

It also contributes to UCX prefix sync workflows when UCX Required is active.

Naming Severity

Off

- Naming problems do not create warnings or blocking issues.

Warning

- Missing prefix or invalid characters create warnings.

Blocking

- Missing prefix or invalid characters block export.

Sanitize Asset Names

When enabled, Safe Fix can replace invalid name characters with underscores and ensure a safe Maya/node-style name.

If an asset name starts with a digit, the sanitization helper can add a safe prefix.

When disabled, the tool does not sanitize names during fix, although the naming check may still report issues.

Rules Tab - Geometry

Max Polycount

Defines the maximum allowed face count for an asset.

The tool uses Maya polyEvaluate face count across the selected asset transform and optionally descendant meshes.

If the asset polycount exceeds Max Polycount, the result depends on Polycount Severity.

Polycount Severity

Off

- Polycount failures are ignored for export status.

Warning

- Exceeding Max Polycount produces a warning.

Blocking

- Exceeding Max Polycount blocks export.

Freeze Required

When enabled, transform values are checked.

The expected frozen state is

- Translate X/Y/Z approximately 0.
- Rotate X/Y/Z approximately 0.
- Scale X/Y/Z approximately 1.

If the object is not frozen, the result depends on Freeze Severity.

Freeze Severity

Off

- Freeze failure is ignored for export status.

Warning

- Non-frozen transforms create a warning.

Blocking

- Non-frozen transforms block export.

Clean History Required

When enabled, the tool checks for construction history beyond ignored Maya utility/history types.

If history is found, the result depends on History Severity.

History Severity

Off

- Construction history does not affect export status.

Warning

- Construction history creates a warning.

Blocking

- Construction history blocks export.

Zero Thickness Warning

When enabled, the bounding box width, height and depth are checked against Zero Thickness Tolerance.

If any axis is equal to or below the tolerance, the tool reports a near-zero axis.

This is useful for identifying planes, collapsed geometry or accidental flat meshes.

Zero Thickness Tolerance

Default

```
0.001 cm
```

Any bounding-box axis at or below this value is treated as near-zero thickness.

Zero Thickness Severity

Off

- Near-zero thickness does not affect export status.

Warning

- Near-zero thickness creates a warning.

Blocking

- Near-zero thickness blocks export.

Rules Tab - Dimensions

Enable Dimension Check

When enabled, the asset bounding-box size is compared against expected width, height and depth.

When disabled, dimension check is skipped.

Expected Width / Height / Depth

These values define the target asset size in centimeters.

Maya Y is treated as height.

Width, height and depth are based on the world bounding box.

Tolerance

Defines how far the actual dimension can differ from the expected dimension.

If all deltas are within tolerance, the dimension check passes.

If any delta is greater than tolerance, the dimension check fails.

Dimension Severity

Off

- Dimension mismatch does not affect export status.

Warning

- Dimension mismatch creates a warning.

Blocking

- Dimension mismatch blocks export.

Important

Safe Fix does not resize geometry automatically. Dimension failures require manual resizing or profile adjustment.

Rules Tab - Pivot

Enable Pivot Check

When enabled, the current rotate pivot is compared to the configured Pivot Target.

When disabled, pivot validation is skipped.

Pivot Target

Available targets include

- Center
- Bottom Center
- Bottom Front Center

- Bottom Back Center
- Bottom Left Center
- Bottom Right Center
- Bottom Front Left
- Bottom Front Right
- Bottom Back Left
- Bottom Back Right
- Front Center
- Back Center
- Left Center
- Right Center
- Top Center
- Top Front Center
- Top Back Center
- Top Left Center
- Top Right Center
- Top Front Left
- Top Front Right
- Top Back Left
- Top Back Right

The tool calculates these positions from the asset world bounding box and the selected Front Axis.

Changing Front Axis can change the meaning of Front, Back, Left and Right.

Tolerance

Defines allowed pivot difference in centimeters.

If the current pivot is within tolerance on X, Y and Z, the pivot check passes.

Pivot Severity

Off

- Pivot mismatch does not affect export status.

Warning

- Pivot mismatch creates a warning.

Blocking

- Pivot mismatch blocks export.

Pivot Fix Behavior

If Move Pivot To Target is enabled in Prep / Fix and the pivot check is enabled, Safe Fix moves the asset pivot to the configured Pivot Target.

The geometry is not moved by the pivot fix. Only the pivot position is changed.

Rules Tab - Grid

Enable Grid Check

When enabled, the tool checks grid readiness using Grid Step.

When disabled, grid validation is skipped.

Grid Step

Default

```
10 cm
```

The grid check uses this step for bounds snapping and size multiple checks.

Check Bounds Snap

When enabled, the tool checks whether these world bounding-box values are multiples of Grid Step:

- minX
- minY
- minZ
- maxX
- maxY
- maxZ

If any are not multiples, bounds snap check fails.

Check Size Multiple

When enabled, the tool checks whether width, height and depth are multiples of Grid Step.

If size is not a multiple of Grid Step, moving the object cannot fix the size. The asset must be resized manually or the profile should be adjusted.

Grid Severity

Off

- Grid failures do not affect export status.

Warning

- Grid failures create warnings.

Blocking

- Grid failures block export.

Grid Fix Behavior

Safe Fix can move the asset if one of these is enabled

- Snap Bounds Min To Grid
- Snap Pivot To Grid

Snap Bounds Min To Grid moves the asset so bounding box minX/minY/minZ align to the nearest grid step.

Snap Pivot To Grid moves the asset so the pivot position aligns to the nearest grid step.

The tool does not resize geometry.

Rules Tab - Advanced UNREAL Readiness

Enable LOD Readiness Check

When enabled, the tool looks for LOD naming patterns based on the selected asset name.

Example

SM_Rock_LOD0 SM_Rock_LOD1 SM_Rock_LOD2

The tool checks

- Whether an LOD group exists.
- Whether LOD0 exists when an LOD group is detected.
- Whether polycount decreases or stays equal across LOD levels.
- Whether LOD pivots are consistent.

If no LOD naming is detected, the check passes because single-mesh workflow is valid.

LOD Readiness Severity

Off

- LOD issues do not affect export status.

Warning

- LOD issues create warnings.

Blocking

- LOD issues block export.

Enable Socket / Locator Readiness

When enabled, the tool searches descendants of the selected asset for socket-style transforms or locator nodes.

Detected socket candidates include

- Names starting with SOCKET_
- Names starting with Socket_
- Transform nodes with locator shapes

The tool checks

- Socket presence.

- SOCKET_* naming convention.
- Locked transform attributes on socket nodes.

No sockets found is valid for simple static meshes.

Socket / Locator Severity

Off

- Socket issues do not affect export status.

Warning

- Socket issues create warnings.

Blocking

- Socket issues block export.

Enable Material Slot Readiness

When enabled, the tool gathers materials connected through shading engines and checks material slot quality.

The tool checks

- Whether any material assignments exist.
- Whether material slot count exceeds Max Material Slots.
- Whether default Maya materials are detected.
- Whether material names use configured prefixes.

Material Prefixes

Default

M_,MI_,MAT_

Material names must start with one of these prefixes to pass the naming check.

Examples that pass with default prefixes

- M_Wood
- MI_Table_Wood
- MAT_Chair_Fabric

Example that fails

- standardSurface1

Changing this list changes which material names are accepted.

Max Material Slots

Default

8

If the number of detected material slots is greater than this value, the Material Slot Count check fails.

Set to a higher number for assets that intentionally use many materials.

Material Slot Severity

Off

- Material slot issues do not affect export status.

Warning

- Material slot issues create warnings.

Blocking

- Material slot issues block export.

Enable UV Readiness Check

When enabled, the tool checks UV readiness.

The tool checks

- UV sets per shape.
- Total UV coordinate count.
- Sampled UV coordinates outside the 0-1 range.
- Required lightmap UV set if Require Lightmap UV is enabled.

Important

UV overlap calculation is advisory/manual in v1.0.8. The tool does not provide full UV shell overlap solving.

Require Lightmap UV

When enabled, the tool requires at least one UV set matching the configured Lightmap UV Names.

When disabled, the lightmap UV check passes even if no lightmap UV set is found.

Lightmap UV Names

Default

lightmap,Lightmap,UV1,uv1,map2

The tool searches UV set names against this comma-separated list.

If Require Lightmap UV is enabled and no UV set matches one of these names, the asset receives a lightmap UV warning or blocking issue depending on UV Readiness Severity.

UV Sample Limit

Default

2000

The tool samples up to this many UV coordinates per shape for the 0-1 range check.

Higher values inspect more UV coordinates but can be slower on dense meshes.

UV Readiness Severity

Off

- UV issues do not affect export status.

Warning

- UV issues create warnings.

Blocking

- UV issues block export.

Prep / Fix Tab - Safe Fix Options

Duplicate Backup Before Fix

When enabled, Safe Fix creates a duplicate backup before modifying the selected asset.

Backups are named using the asset name plus an M2U backup suffix.

This is recommended for safer workflows.

Hide Backups

When enabled, backup duplicates are hidden after creation.

When disabled, backups remain visible in the Maya scene.

Rename / Add Prefix If Missing

When enabled, Safe Fix can rename the base asset to satisfy the naming rule.

It can

- Add the Required Prefix if missing.
- Use sanitized naming if Sanitize Asset Names is enabled.
- Avoid duplicate names by creating a unique final name.

Example

Chair_01 becomes SM_Chair_01 when Required Prefix is SM_.

Sync UCX Names To Base Prefix When UCX Required

When enabled, matching UCX collision names are synchronized if the base mesh is renamed during Safe Fix.

This only runs when

- Rename / Add Prefix If Missing is enabled.
- Collision Check is enabled.
- Collision Requirement is UCX required.
- Collision Match Mode is Base asset name.

Example

Before Safe Fix

Chair_01 UCX_Chair_01 UCX_Chair_01_02

Required Prefix

```
SM_
```

After Safe Fix

SM_Chair_01 UCX_SM_Chair_01 UCX_SM_Chair_01_02

Multi-part UCX suffixes are preserved.

If Collision Match Mode is Exact custom target name, the sync is intentionally skipped to avoid changing artist-specified custom target workflows.

Freeze Transform

When enabled, Safe Fix applies Maya makeIdentity to freeze translate, rotate and scale when a freeze issue is detected.

It does not run if the transform already appears frozen.

Delete Construction History

When enabled, Safe Fix deletes construction history when history is detected.

This can remove procedural history and should be used with backups enabled.

Move Pivot To Target

When enabled, Safe Fix moves the pivot to the configured Pivot Target.

Pivot Check must also be enabled for the target settings to be meaningful.

This does not move the mesh geometry.

Unlock Transform Attributes

When enabled, Safe Fix unlocks transform-related attributes before applying fixes.

Affected attributes can include

- translateX/Y/Z
- rotateX/Y/Z
- scaleX/Y/Z
- visibility

This is disabled by default because locked attributes may be intentional in some scenes.

Make Selected Assets Visible

When enabled, Safe Fix attempts to set selected asset visibility to true.

This is disabled by default because hidden visibility may be intentional.

Snap Bounds Min To Grid

When enabled, Safe Fix moves the asset so its bounding box minimum point aligns to the nearest grid step.

This changes asset position but does not resize geometry.

Snap Pivot To Grid

When enabled, Safe Fix moves the asset so its pivot aligns to the nearest grid step.

This changes asset position but does not resize geometry.

If both Snap Bounds Min To Grid and Snap Pivot To Grid are enabled, the current implementation prioritizes bounds-min snapping first.

Prep / Fix Tab - Preview / Artist-controlled Helpers

Create Pivot Ghost Preview

Creates a locator at the current asset's calculated target pivot position.

The locator name starts with

M2U_PIVOT_PREVIEW_

This lets artists see where the pivot will be moved before applying Safe Fix.

It does not modify the asset pivot.

Delete Pivot Preview Locators

Deletes only M2U-created pivot preview locator transforms whose names start with:

M2U_PIVOT_PREVIEW_

It does not delete mesh assets, UCX meshes, backups, or user-created non-M2U locators.

Create Simple Box UCX

Creates a simple box-shaped UCX collision mesh based on the selected asset's bounding box.

The created collision mesh is named

UCX_<BaseAssetName>

or a unique variant if that name already exists.

This is an artist-controlled helper for quick basic collision. It is not a replacement for carefully authored gameplay collision when accurate collision is required.

Collision Tab - Collision Rules

Enable Collision Check

When enabled, the tool evaluates UCX collision according to the selected requirement.

When disabled, collision validation is skipped.

Requirement

Available values

Off

- Collision check is disabled.

UCX required

- Matching UCX collision must exist.
- If no match is found, the asset receives an issue based on Collision Severity.
- At export time, the tool searches again. If no matching UCX exists, export fails for that asset to prevent base-only export.

UCX optional

- Matching UCX collision is used if found.
- Missing UCX is not a failure.
- Existing matching UCX can still be validated if Validate Matched UCX Meshes is enabled.

No custom collision allowed

- Matching UCX collision is considered a failure.
- Use this when the target workflow should not export custom UCX meshes.

Match Mode

Base asset name

- The target collision name is derived from the base mesh name.
- Recommended for normal Unreal UCX workflows.

Exact custom target name

- The target collision name is taken from Custom Target Name.
- Use this when UCX meshes intentionally target a different name.
- UCX prefix sync is disabled in this mode for safety.

Custom Target Name

Used only when Match Mode is Exact custom target name.

The tool searches for

UCX_<CustomTargetName> UCX_<CustomTargetName>_01 UCX_<CustomTargetName>_02

when multiple parts are accepted.

Accept Multiple UCX Parts

When enabled, the tool accepts

UCX_SM_Table_01 UCX_SM_Table_01_01 UCX_SM_Table_01_02

When disabled, the tool only accepts the exact UCX target name without additional numbered suffixes.

Validate Matched UCX Meshes

When enabled, matched UCX meshes are checked separately.

UCX validation includes

- Has mesh shape.

- Freeze transform.
- Construction history.
- Polycount.
- Zero thickness.
- Visibility in hierarchy.
- Locked transform attributes.

UCX Mesh Validation Severity

Off

- UCX mesh validation failures do not affect export status.

Warning

- UCX mesh validation failures create warnings.

Blocking

- UCX mesh validation failures block export.

Collision Severity

Controls how main collision rule failures affect the asset.

Examples

- UCX required but no match found.
- Custom UCX found when No custom collision allowed is active.

Off

- Collision rule failures do not affect export status.

Warning

- Collision rule failures create warnings.

Blocking

- Collision rule failures block export.

Collision Tab - Collision Visualizer

Select Asset + Matching UCX

Selects the currently highlighted result asset and its matching UCX collision meshes in Maya.

Use this to quickly inspect collision pairing in the viewport or Outliner.

Isolate Asset + UCX

Attempts to isolate the current asset and matching UCX meshes in the active model panel.

This depends on Maya viewport/model panel state.

Colorize UCX Preview

Assigns a preview material to matching UCX meshes so they are visually easier to identify.

Before assignment, the tool remembers the UCX shading assignments for the current Maya session.

Reset UCX Preview Materials

Restores UCX preview material assignments remembered during the current Maya session.

Important

This restore memory is session-based. If Maya is closed after colorizing, the tool cannot restore previous assignments from memory.

Presets

Marketplace Static Mesh

General-purpose default preset for Unreal-ready marketplace/static mesh assets.

Typical settings

- Required Prefix: SM_
- Collision Requirement: UCX optional
- Dimension Check: disabled
- Grid Step: 10 cm
- Max Polycount: 12000
- Pivot Target: Bottom Center
- Lightmap UV Required: disabled

Best for

- Props.
- Furniture.
- General static meshes.
- Marketplace-style asset packs.

Modular Wall Kit

Stricter preset for modular environment pieces.

Typical settings

- Collision Requirement: UCX required
- Collision Severity: blocking
- Grid Step: 50 cm
- Grid Severity: blocking
- Dimension Check: enabled
- Expected Size: 400 x 300 x 20 cm
- Pivot Target: Bottom Front Center
- Pivot Severity: blocking

Best for

- Modular wall pieces.
- Kitbash wall systems.
- Grid-snapped modular environments.

Furniture / Prop

General prop preset with more flexible polycount.

Typical settings

- Collision Requirement: UCX optional
- Max Polycount: 20000
- Pivot Target: Bottom Center
- Dimension Check: disabled
- Grid Step: 10 cm

Best for

- Chairs.
- Tables.
- Decoration props.
- Furniture sets.

Collision Strict

Strict collision-focused preset.

Typical settings

- Collision Requirement: UCX required
- Collision Severity: blocking
- Validate Matched UCX Meshes: enabled
- UCX Mesh Validation Severity: blocking
- Max Polycount: 10000

Best for

- Game assets that must ship with custom collision.
- Modular pieces with required collision.
- QA-controlled export passes.

Nanite Candidate Check

Preflight preset for high-poly static mesh candidates.

Typical settings

- Max Polycount: 500000
- Polycount Severity: warning
- Collision Requirement: UCX optional

- Dimension Check: disabled
- LOD Readiness Check: disabled
- Lightmap UV Required: disabled

Best for

- High-poly Unreal static mesh candidates.
- Nanite-oriented review passes.

Important

This preset does not enable Nanite in Unreal automatically. It is a Maya-side preflight preset.

Reports

HTML Report

Artist-friendly visual QA report.

Includes

- Total, Ready, Warning, Blocked and Exported counts.
- Per-asset M2U Score.
- Asset status.
- Export status.
- Polycount.
- Profile.
- Included UCX count.
- Issues.
- Fix Plan.
- Fixed Actions.
- Collision Matches.
- Collision Target Candidates.
- Core Checks.
- LOD / Material Slot / UV / Socket Readiness.

JSON Report

Technical report for debugging or pipeline integration.

Includes full structured data

- Settings.
- Export summary.
- Per-asset results.
- Warnings and blocking issues.
- Fix plans and fixed actions.

- Collision match data.
- Validation data.
- Export status.

Report Error Isolation

Report writing is isolated from FBX export success.

If an FBX exports successfully but a report writer fails, the successful FBX export should not be turned into a false export failure. Report errors are collected in the export summary.

FBX Export Details

The tool loads the Maya FBX plugin if needed.

During export, the tool selects

- The base asset transform.
- Matching UCX collision transforms that exist and are allowed by the collision settings.

FBX export settings requested through MEL include

- FBXResetExport
- Export smoothing groups enabled
- Export tangents enabled
- Export smooth mesh enabled
- Export instances disabled
- Export triangulation based on Triangulate FBX Export
- Up axis set to Y

Each exported asset receives its own FBX file.

UCX Export-time Recheck

Before exporting each asset, the tool searches for UCX matches again.

This prevents stale results when

- The user created UCX after the first analysis.
- The user renamed UCX after analysis.
- Safe Fix renamed base meshes and synced UCX names.

If UCX Required is active and no matching UCX exists at export time, the export fails for that asset with a clear message listing searched target candidates.

Limitations And Intentional Boundaries

M2U Pipeline Suite is intentionally conservative.

It does not

- Automatically resize geometry.
- Automatically remodel geometry.
- Solve UV overlaps.
- Create advanced convex decomposition collision.
- Create Unreal materials or material instances.
- Fully configure Nanite settings in Unreal.
- Fully configure Unreal LOD groups or screen sizes.
- Guarantee Unreal socket setup inside static mesh assets.

It does

- Detect common pipeline problems.
- Apply safe artist-controlled cleanup operations.
- Export base assets with matching UCX collision.
- Produce detailed reports.
- Generate an Unreal Python import helper.

Recommended Production Workflow

For a normal static mesh asset

1. Name the base mesh using the project prefix, for example SM_Table_01.
2. Name matching UCX collision as UCX_SM_Table_01 or UCX_SM_Table_01_01.
3. Assign material names using M_, MI_ or MAT_ prefixes.
4. Select the base mesh transform only.
5. Apply Marketplace Static Mesh or Furniture / Prop preset.
6. Analyze Selected Assets.
7. Review warnings.
8. Apply Safe Fixes if the fix plan is acceptable.
9. Revalidate.
10. Export FBX + Reports.
11. Review M2U_Pipeline_Report.html.
12. Import FBX into Unreal manually or run the generated Unreal import script.

For a strict modular asset

1. Use Modular Wall Kit or Collision Strict preset.
2. Confirm dimensions match the expected modular size.
3. Confirm grid size and bounds pass.
4. Confirm UCX collision exists and validates.
5. Treat blocked status as a required fix before export.

19. Appendix: Production Workflow Examples

Normal static mesh asset

- Name the base mesh using the project prefix, for example SM_Table_01.
- Name matching UCX collision as UCX_SM_Table_01 or UCX_SM_Table_01_01.
- Assign material names using M_, MI_ or MAT_ prefixes.
- Select the base mesh transform only.
- Apply Marketplace Static Mesh or Furniture / Prop preset.
- Analyze Selected Assets.
- Review warnings and the detailed result panel.
- Apply Safe Fixes only if the fix plan is acceptable.
- Revalidate.
- Export FBX + Reports.
- Review M2U_Pipeline_Report.html.
- Import FBX manually into Unreal or run the generated Unreal import script.

Strict modular asset

- Use Modular Wall Kit or Collision Strict preset.
- Confirm expected dimensions, grid size and pivot target match the modular design.
- Confirm UCX collision exists and validates.
- Treat Blocked status as a required fix before export.
- Use the HTML report as the final QA record before delivery.

20. Document Basis

This PDF was assembled from the provided public README, the detailed Functions and Options Guide, and the supplied M2U Pipeline Suite screenshots. The documentation text has been reorganized into a professional user-guide and function-reference format for public release or client-facing delivery.